Exercise sheet 3

Linear Optimization (311.160), WS 2026/27

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Exercise 3.1

Maximize $2x_1 + 3x_2 + x_3$ *subject to*

$$x_1 + 2x_2 + x_3 + x_4 = 4, \qquad 2x_1 + x_2 + 2x_3 + x_5 = 5, \qquad x_1, x_2, x_3, x_4, x_5 \geq 0.$$

1. Determine A and b. 2. Choose a basis and its nonbasis. 3. Write $A_B x_B = b - A_N x_N$. 4. Solve for x_B in terms of x_N.

1. Matrix and right hand side

With $x = (x_1, x_2, x_3, x_4, x_5)^T$ the system is $Ax = b$, where

$$\boxed{A = \begin{pmatrix} 1 & 2 & 1 & 1 & 0 \\ 2 & 1 & 2 & 0 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 4 \\ 5 \end{pmatrix}.}$$

Check by matrix multiplication:

$$\begin{aligned} Ax &= \begin{pmatrix} 1x_1 + 2x_2 + 1x_3 + 1x_4 + 0x_5 \\ 2x_1 + 1x_2 + 2x_3 + 0x_4 + 1x_5 \end{pmatrix} && \text{row times column} \\ &= \begin{pmatrix} x_1 + 2x_2 + x_3 + x_4 \\ 2x_1 + x_2 + 2x_3 + x_5 \end{pmatrix} && \text{remove zero terms} \end{aligned}$$

These are the two given left hand sides.

2. Basis and nonbasis

Choose the ordered index sets $B = (4, 5)$ and $N = (1, 2, 3)$. Then

$$A_B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad A_N = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \end{pmatrix}, \quad x_B = \begin{pmatrix} x_4 \\ x_5 \end{pmatrix}, \quad x_N = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

$$\begin{aligned} \det(A_B) &= 1 \cdot 1 - 0 \cdot 0 && \text{determinant of a } 2 \times 2 \text{ matrix} \\ &= 1 - 0 && \text{multiply} \\ &= 1 && \text{subtract} \end{aligned}$$

Thus the two selected columns are independent and form a basis.

$$\boxed{B = (4, 5), \qquad N = (1, 2, 3).}$$

3. Split system

$$\begin{aligned} A_B x_B + A_N x_N &= b && \text{partition the columns of } A \\ A_B x_B &= b - A_N x_N && \text{subtract } A_N x_N \end{aligned}$$

$$\boxed{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 4 \\ 5 \end{pmatrix} - \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.}$$

Check: adding $A_N x_N$ recovers $Ax = b$ with the same column order.

4. Solve for the basic variables

$$
\begin{aligned}
x_B &= A_B^{-1}(b - A_N x_N) && \text{multiply by } A_B^{-1} \\
&= I_2(b - A_N x_N) && A_B \text{ is the identity} \\
&= b - A_N x_N && \text{identity multiplication} \\
&= \begin{pmatrix} 4 \\ 5 \end{pmatrix} - \begin{pmatrix} x_1 + 2x_2 + x_3 \\ 2x_1 + x_2 + 2x_3 \end{pmatrix} && \text{matrix multiplication} \\
&= \begin{pmatrix} 4 - x_1 - 2x_2 - x_3 \\ 5 - 2x_1 - x_2 - 2x_3 \end{pmatrix} && \text{subtract componentwise}
\end{aligned}
$$

$$
\boxed{x_4 = 4 - x_1 - 2x_2 - x_3, \qquad x_5 = 5 - 2x_1 - x_2 - 2x_3.}
$$

Check for arbitrary x_N:

$$
\begin{aligned}
x_1 + 2x_2 + x_3 + x_4 &= x_1 + 2x_2 + x_3 + (4 - x_1 - 2x_2 - x_3) && \text{substitute } x_4 \\
&= 4 && \text{cancel opposite terms} \\
2x_1 + x_2 + 2x_3 + x_5 &= 2x_1 + x_2 + 2x_3 + (5 - 2x_1 - x_2 - 2x_3) && \text{substitute } x_5 \\
&= 5 && \text{cancel opposite terms}
\end{aligned}
$$

The choice must satisfy $x_N \geq 0$ and $x_B \geq 0$. Setting $x_N = 0$ gives

$$
\begin{aligned}
x_B &= \begin{pmatrix} 4 - 0 - 2 \cdot 0 - 0 \\ 5 - 2 \cdot 0 - 0 - 2 \cdot 0 \end{pmatrix} && \text{substitute the nonbasic zeros} \\
&= \begin{pmatrix} 4 \\ 5 \end{pmatrix} && \text{evaluate zero terms}
\end{aligned}
$$

Both entries are nonnegative, so $(0, 0, 0, 4, 5)^T$ is basic feasible.

Exercise 3.2

Maximize $z = x_1 - 2x_2 - 3x_3$ *subject to*

$$
-x_1 + 2x_2 + 3x_3 \leq 3, \qquad x_1 - x_2 - 2x_3 \leq 2, \qquad 2x_1 + x_2 + x_3 \leq 7, \qquad x_1, x_2, x_3 \geq 0.
$$

1. Introduce x_4, x_5, x_6 *for standard form and compute the slacks and objective at* $x_1 = x_2 = x_3 = 0$. *2. Solve for* x_4, x_5, x_6. *How far can only* x_1 *increase while all variables remain nonnegative? 3. Could increasing* x_2 *or* x_3 *help solve the maximization problem?*

1. Standard form and initial values

For each upper bound add a nonnegative slack variable:

$$
\boxed{\begin{aligned}
&\max\ z = x_1 - 2x_2 - 3x_3, \\
&-x_1 + 2x_2 + 3x_3 + x_4 = 3, \\
&x_1 - x_2 - 2x_3 + x_5 = 2, \\
&2x_1 + x_2 + x_3 + x_6 = 7, \\
&x_1, x_2, x_3, x_4, x_5, x_6 \geq 0.
\end{aligned}}
$$

At the origin:

$$
-0 + 2 \cdot 0 + 3 \cdot 0 + x_4 = 3 \qquad \text{substitute } x_1, x_2, x_3
$$

$$
\begin{aligned}
x_4 &= 3 && \text{evaluate zero terms}\\
0 - 0 - 2\cdot 0 + x_5 &= 2 && \text{substitute } x_1, x_2, x_3\\
x_5 &= 2 && \text{evaluate zero terms}\\
2\cdot 0 + 0 + 0 + x_6 &= 7 && \text{substitute } x_1, x_2, x_3\\
x_6 &= 7 && \text{evaluate zero terms}\\
z &= 0 - 2\cdot 0 - 3\cdot 0 && \text{substitute in the objective}\\
&= 0 && \text{evaluate zero terms}
\end{aligned}
$$

$$\boxed{(x_4, x_5, x_6) = (3, 2, 7), \qquad z = 0.}$$

Check: the original three left hand sides are zero at the origin, hence satisfy their positive upper bounds.

2. Initial dictionary and increase of x_1

$$
\begin{aligned}
-x_1 + 2x_2 + 3x_3 + x_4 &= 3 && \text{first equation}\\
2x_2 + 3x_3 + x_4 &= 3 + x_1 && \text{add } x_1\\
3x_3 + x_4 &= 3 + x_1 - 2x_2 && \text{subtract } 2x_2\\
x_4 &= 3 + x_1 - 2x_2 - 3x_3 && \text{subtract } 3x_3
\end{aligned}
$$

$$
\begin{aligned}
x_1 - x_2 - 2x_3 + x_5 &= 2 && \text{second equation}\\
-x_2 - 2x_3 + x_5 &= 2 - x_1 && \text{subtract } x_1\\
-2x_3 + x_5 &= 2 - x_1 + x_2 && \text{add } x_2\\
x_5 &= 2 - x_1 + x_2 + 2x_3 && \text{add } 2x_3
\end{aligned}
$$

$$
\begin{aligned}
2x_1 + x_2 + x_3 + x_6 &= 7 && \text{third equation}\\
x_2 + x_3 + x_6 &= 7 - 2x_1 && \text{subtract } 2x_1\\
x_3 + x_6 &= 7 - 2x_1 - x_2 && \text{subtract } x_2\\
x_6 &= 7 - 2x_1 - x_2 - x_3 && \text{subtract } x_3
\end{aligned}
$$

$$\boxed{x_4 = 3 + x_1 - 2x_2 - 3x_3, \quad x_5 = 2 - x_1 + x_2 + 2x_3, \quad x_6 = 7 - 2x_1 - x_2 - x_3.}$$

Keep $x_2 = x_3 = 0$ and set $x_1 = t \geq 0$:

$$
\begin{aligned}
x_4 &= 3 + t - 2\cdot 0 - 3\cdot 0 && \text{substitute}\\
&= 3 + t && \text{evaluate zero terms}\\
x_5 &= 2 - t + 0 + 2\cdot 0 && \text{substitute}\\
&= 2 - t && \text{evaluate zero terms}\\
x_6 &= 7 - 2t - 0 - 0 && \text{substitute}\\
&= 7 - 2t && \text{evaluate zero terms}
\end{aligned}
$$

The first slack is nonnegative for every $t \geq 0$. For the other two:

$$
\begin{aligned}
2 - t &\geq 0 && \text{nonnegativity of } x_5\\
2 &\geq t && \text{add } t\\
7 - 2t &\geq 0 && \text{nonnegativity of } x_6\\
7 &\geq 2t && \text{add } 2t
\end{aligned}
$$

$$\frac{7}{2} \geq t \qquad \text{divide by positive 2}$$

$$\boxed{0 \leq x_1 \leq 2; \qquad x_1 \text{ can increase by 2, and } x_5 \text{ leaves the basis.}}$$

At the endpoint:

$$\begin{aligned} x_4 &= 3 + 2 && \text{substitute } t = 2 \\ &= 5 && \text{add} \\ x_5 &= 2 - 2 && \text{substitute } t = 2 \\ &= 0 && \text{subtract} \\ x_6 &= 7 - 2 \cdot 2 && \text{substitute } t = 2 \\ &= 7 - 4 && \text{multiply} \\ &= 3 && \text{subtract} \end{aligned}$$

Check in the original inequalities:

$$\begin{aligned} -x_1 + 2x_2 + 3x_3 &= -2 + 2 \cdot 0 + 3 \cdot 0 && \text{substitute } (2, 0, 0) \\ &= -2 && \text{evaluate zero terms} \\ -2 &\leq 3 && \text{first bound} \\ x_1 - x_2 - 2x_3 &= 2 - 0 - 2 \cdot 0 && \text{substitute} \\ &= 2 && \text{evaluate zero terms} \\ 2 &\leq 2 && \text{second bound} \\ 2x_1 + x_2 + x_3 &= 2 \cdot 2 + 0 + 0 && \text{substitute} \\ &= 4 && \text{multiply and remove zeros} \\ 4 &\leq 7 && \text{third bound} \end{aligned}$$

3. Other entering variables

At the initial basic solution, increasing only $x_2 = t$ gives

$$\begin{aligned} z &= 0 - 2t - 3 \cdot 0 && \text{set } x_1 \text{ and } x_3 \text{ to zero} \\ &= -2t && \text{remove zeros} \\ z &< 0 && t \text{ positive} \end{aligned}$$

Increasing only $x_3 = t$ instead gives

$$\begin{aligned} z &= 0 - 2 \cdot 0 - 3t && \text{set } x_1 \text{ and } x_2 \text{ to zero} \\ &= -3t && \text{remove zeros} \\ z &< 0 && t \text{ positive} \end{aligned}$$

$$\boxed{\text{Neither } x_2 \text{ nor } x_3 \text{ gives an improving initial step.}}$$

Check: both slopes, -2 and -3, are strictly negative, whereas the initial objective is zero.

Exercise 3.3

Continue with Exercise 3.2. 4. Express x_1, x_4, x_6 in terms of x_2, x_3, x_5. 5. Express the objective in these variables and explain why the current basic solution is optimal. 6. State the optimal solution and optimal objective value.

4. New dictionary

Solve the second equation for the new basic variable x_1:

$$\begin{aligned} x_5 &= 2 - x_1 + x_2 + 2x_3 && \text{previous dictionary} \\ x_1 + x_5 &= 2 + x_2 + 2x_3 && \text{add } x_1 \\ x_1 &= 2 + x_2 + 2x_3 - x_5 && \text{subtract } x_5 \end{aligned}$$

$$\begin{aligned} x_4 &= 3 + x_1 - 2x_2 - 3x_3 && \text{previous dictionary} \\ &= 3 + (2 + x_2 + 2x_3 - x_5) - 2x_2 - 3x_3 && \text{substitute } x_1 \\ &= (3+2) + (1-2)x_2 + (2-3)x_3 - x_5 && \text{collect equal terms} \\ &= 5 - x_2 - x_3 - x_5 && \text{evaluate coefficients} \end{aligned}$$

$$\begin{aligned} x_6 &= 7 - 2x_1 - x_2 - x_3 && \text{previous dictionary} \\ &= 7 - 2(2 + x_2 + 2x_3 - x_5) - x_2 - x_3 && \text{substitute } x_1 \\ &= 7 - 4 - 2x_2 - 4x_3 + 2x_5 - x_2 - x_3 && \text{distribute } -2 \\ &= (7-4) + (-2-1)x_2 + (-4-1)x_3 + 2x_5 && \text{collect equal terms} \\ &= 3 - 3x_2 - 5x_3 + 2x_5 && \text{evaluate coefficients} \end{aligned}$$

$$\boxed{\begin{aligned} x_1 &= 2 + x_2 + 2x_3 - x_5, \\ x_4 &= 5 - x_2 - x_3 - x_5, \\ x_6 &= 3 - 3x_2 - 5x_3 + 2x_5. \end{aligned}}$$

The new basis is $B = (1, 4, 6)$ and the nonbasis is $N = (2, 3, 5)$. Check the identities in all three original equations:

$$\begin{aligned} -x_1 + 2x_2 + 3x_3 + x_4 &= -(2 + x_2 + 2x_3 - x_5) + 2x_2 + 3x_3 + (5 - x_2 - x_3 - x_5) && \text{substitute} \\ &= -2 - x_2 - 2x_3 + x_5 + 2x_2 + 3x_3 + 5 - x_2 - x_3 - x_5 && \text{remove parentheses} \\ &= (-2+5) + (-1+2-1)x_2 + (-2+3-1)x_3 + (1-1)x_5 && \text{collect terms} \\ &= 3 && \text{evaluate coefficients} \\ x_1 - x_2 - 2x_3 + x_5 &= (2 + x_2 + 2x_3 - x_5) - x_2 - 2x_3 + x_5 && \text{substitute} \\ &= 2 && \text{cancel opposite terms} \\ 2x_1 + x_2 + x_3 + x_6 &= 2(2 + x_2 + 2x_3 - x_5) + x_2 + x_3 + (3 - 3x_2 - 5x_3 + 2x_5) && \text{substitute} \\ &= 4 + 2x_2 + 4x_3 - 2x_5 + x_2 + x_3 + 3 - 3x_2 - 5x_3 + 2x_5 && \text{distribute } 2 \\ &= (4+3) + (2+1-3)x_2 + (4+1-5)x_3 + (-2+2)x_5 && \text{collect terms} \\ &= 7 && \text{evaluate coefficients} \end{aligned}$$

5. Objective and optimality

$$\begin{aligned} z &= x_1 - 2x_2 - 3x_3 && \text{original objective} \\ &= (2 + x_2 + 2x_3 - x_5) - 2x_2 - 3x_3 && \text{substitute } x_1 \\ &= 2 + (1-2)x_2 + (2-3)x_3 - x_5 && \text{collect terms} \\ &= 2 - x_2 - x_3 - x_5 && \text{evaluate coefficients} \\ z &\le 2 && \text{all nonbasic variables nonnegative} \end{aligned}$$

At $x_2 = x_3 = x_5 = 0$ the dictionary gives:

$$x_1 = 2 + 0 + 2 \cdot 0 - 0 \qquad \text{substitute}$$

$$
\begin{aligned}
&= 2 && \text{evaluate zeros}\\
x_4 &= 5 - 0 - 0 - 0 && \text{substitute}\\
&= 5 && \text{evaluate zeros}\\
x_6 &= 3 - 3 \cdot 0 - 5 \cdot 0 + 2 \cdot 0 && \text{substitute}\\
&= 3 && \text{evaluate zeros}\\
z &= 2 - 0 - 0 - 0 && \text{substitute}\\
&= 2 && \text{evaluate zeros}
\end{aligned}
$$

All basic values are nonnegative. The upper bound is attained, so this basic solution is optimal. □

$$\boxed{z = 2 - x_2 - x_3 - x_5 \le 2, \qquad z^* = 2.}$$

6. Optimal solution and check

$$\boxed{(x_1^*, x_2^*, x_3^*) = (2, 0, 0), \qquad (x_4^*, x_5^*, x_6^*) = (5, 0, 3), \qquad z^* = 2.}$$

The full standard-form solution is $(2, 0, 0, 5, 0, 3)^T$. Check:

$$
\begin{aligned}
-x_1 + 2x_2 + 3x_3 + x_4 &= -2 + 2 \cdot 0 + 3 \cdot 0 + 5 && \text{substitute}\\
&= -2 + 5 && \text{evaluate zero terms}\\
&= 3 && \text{add}\\
x_1 - x_2 - 2x_3 + x_5 &= 2 - 0 - 2 \cdot 0 + 0 && \text{substitute}\\
&= 2 && \text{evaluate zero terms}\\
2x_1 + x_2 + x_3 + x_6 &= 2 \cdot 2 + 0 + 0 + 3 && \text{substitute}\\
&= 4 + 3 && \text{multiply and remove zeros}\\
&= 7 && \text{add}\\
z &= 2 - 2 \cdot 0 - 3 \cdot 0 && \text{substitute in original objective}\\
&= 2 && \text{evaluate zero terms}
\end{aligned}
$$

All six variables are nonnegative. Equality in the upper bound requires $x_2 + x_3 + x_5 = 0$. Since the three terms are nonnegative, each must be zero. The dictionary then fixes $x_1 = 2$, so the optimal solution is unique. □